ARENA SCHOOL
www.arenaschool.org

ACAMIS TECHNOLOGY CONFERENCE · OCTOBER 18, 2026

A case study from the INTERNATIONAL SCHOOL OF BEIJING 北京顺义国际学校

From Lab *to Classroom*

Running an AI Student Program Inside a School

THE NEXT 60 MINUTES

From what students built in five days *to what you can run at your school.*

1	**The case**	What ISB students built in five days, and the 20+ AI tools they used
2	**Why it worked**	The research, the three skills, and the teacher's role as coach
3	**Build one, live**	You name a school chore. We build the fix together with Claude Code.
4	**Take it back**	Why teachers need AI too, and what it takes to run this

ABOUT ARENA SCHOOL

Helping students, families, educators, and schools *thrive in the age of AI.*

SELECTED FOR

ANTHROP\C *Startup Program*

OpenAI *Partner Network*

WHAT WE DO

Hands-on programs where middle and high school students make real things with AI, and the teachers and families around them learn to guide that work.

HOW WE TEACH

Every program builds three human skills: *agency, taste, and craft.* Students make every decision; AI helps them build.

WHO WE WORK WITH

- Students, grades 6 to 12
- Teachers and schools
- Parents and families

THE FOUNDING TEAM

Caltech · Harvard · Brown · Tsinghua University · International School of Beijing 北京顺义国际学校 · McKinsey&Company · Goldman Sachs · Y · Microsoft AI · Deloitte. · KIMI

THE QUESTION WE TRY TO ANSWER AT ARENA SCHOOL

When AI can do all of these in seconds, what must our students *still learn to do?*

WHY PARENTS CARE

Parents are asking one practical question: *which skills will outlast the tools?*

From surveying parents in the ISB community and other local communities in mainland China

WHAT PARENTS CARE ABOUT

Which skills last as the tools change?

Will AI do the thinking for my child?

What should my child practice now?

WHAT YOU CAN TELL THEM AFTER TODAY

Three skills outlast any tool

Used well, AI deepens thinking

Real products, in five days

Parents don't see AI as an academic subject, like learning to code.
They see it as a life skill: optional for many jobs today, and likely non-negotiable by the time their children start working.

Staying competitive with AI takes three human skills. *Without them, AI does the thinking.*

01 Agency

Choosing what is worth doing, before anyone assigns it.

WITHOUT IT

AI picks the problem, and students follow.

02 Taste

Judging whether the output is actually good.

WITHOUT IT

Students accept the first answer AI gives.

03 Craft

Finishing the work so people can trust it.

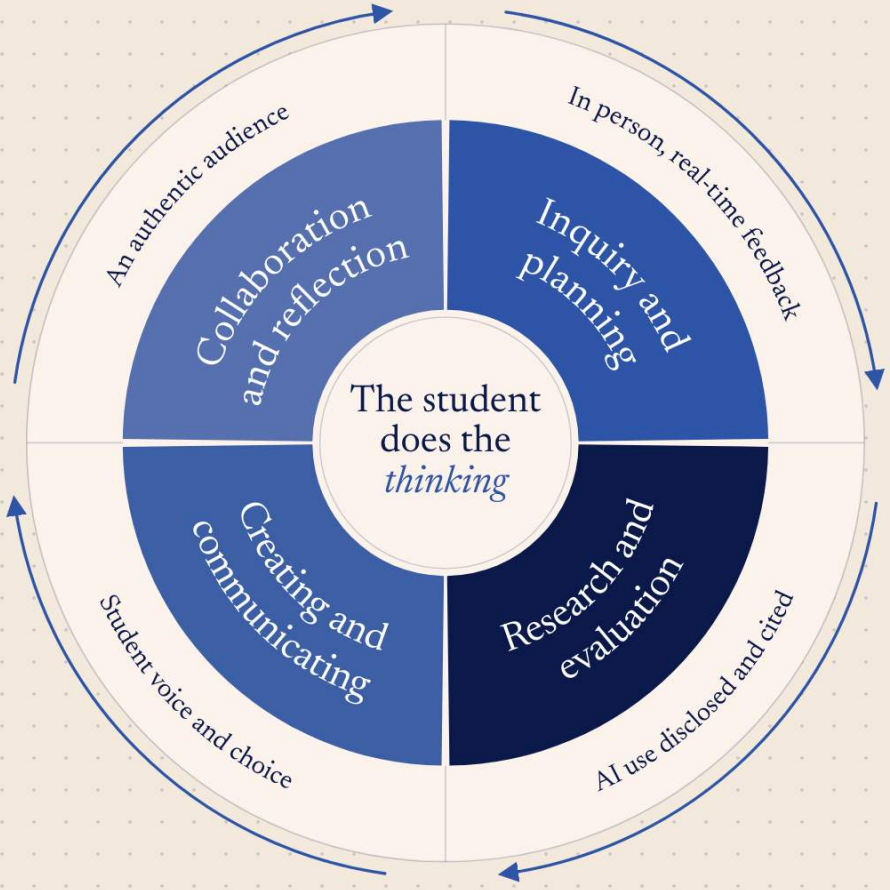
WITHOUT IT

The work stays an unchecked first draft.

This is how we teach students to use AI, *and the lens for everything you'll see today.*

THE ARENA SCHOOL LEARNING FRAMEWORK

Students do the thinking, AI is a tool, *and teachers and peers give the feedback.*



BUILT ON PROJECT-BASED LEARNING

Project-based learning (PBL) gives each strand a real purpose and an authentic audience, and AI speeds up the building.

Informed by PBLWorks' Gold Standard PBL.

AT EVERY CHECK-IN, THE TEACHER ASKS

"How did you check it?"

and responds in real time. AI tools can support that feedback; they don't replace it.

LEARNING OUTCOMES

Across the program, *students will be able to...*

The same outcomes scale from middle school to high school, with more independence and complexity each year.

Inquiry and planning	• Turn a broad interest into a focused, researchable question. • Plan a project with milestones, and adjust the plan based on progress.
Research and evaluation	• Find, evaluate, and cite sources, including checking the accuracy of AI output. • Compare options against criteria, and justify a choice. • Explain how AI tools work, where they fail, and the bias and privacy issues they raise.
Creating and communicating	• Build and test a product, prototype, or solution, deciding which steps to hand to AI and disclosing how they used it. • Structure an argument and write it clearly, in their own words. • Present to an authentic audience, and defend their reasoning under questioning.
Collaboration and reflection	• Work in a team, share roles, and contribute to a shared product. • Give and use critique from teachers and peers to revise their work. • Reflect on their process and on what they would do differently.

HOW WE ASSESS

We assess the process, *not only the product.*

SAMPLE CRITERION · RESEARCH AND EVALUATION

Evaluates sources, including AI output

BEGINNING	DEVELOPING	PROFICIENT	EXTENDING
Accepts sources and AI output as given; cites inconsistently.	Checks some claims against a second source; cites most sources.	Verifies key claims, including AI output, against reliable sources; discloses AI use.	Explains why sources disagree, corrects AI errors, and judges which evidence to trust.

EVIDENCE WE COLLECT

- Process journal
- Drafts
- Recorded defense

IN PRACTICE

Used with students in the ISB AI Founders Program, summer 2026.

PART 1

In five days, ISB students built and launched *working products with AI.*

The program, the week, and what students made

1 The case
What students built

2 Why it worked
The design

3 Build one, live
With the room

4 Take it back
What it takes

THE CASE · ISB AI FOUNDERS PROGRAM, AUGUST 5–9, 2026

An AI builder week for grades 7 to 12: *learn AI skills, start a company, build something real.*



Mentors in the room every day



Every team demos what works

7–12 grades, selected by application

5 days, at ISB in August

0 coding experience required

Students learned 20+ AI tools in five days, *out of 200+ they could choose from.*

200+ AI TOOLS STUDENTS COULD CHOOSE FROM · 20+ THEY USED

RESEARCH

Ask and search

- ChatGPT
- Gemini
- NotebookLM
- Manus
- Mistral
- Claude
- Perplexity
- DeepSeek
- Grok

DESIGN

Logos and art

- Midjourney
- Ideogram
- Krea
- Recraft

BUILD

Apps and sites

- Cursor
- Lovable
- Replit
- v0
- Windsurf

BRAND

Sound and video

- Suno
- ElevenLabs
- CapCut

LAUNCH

Sell and promote

- Stripe
- Pika
- Runway

Using these 20+ AI tools, *students run a "startup" in ten stages, over five days.*

DAY 1

Pick a problem

A one-page plan and a first prototype.

01 Find the idea

02 Validate it

03 Form the team

04 Define it

DAY 2

Go public

Real people try it. Teams fix what confuses them.

05 Design it

06 Build it

07 Ship it live

DAY 3

The rebuild

The best ones outgrow the first plan.

08 Test and iterate

DAY 4

Polish and price

A name, a landing page, a real price.

09 Brand it

DAY 5

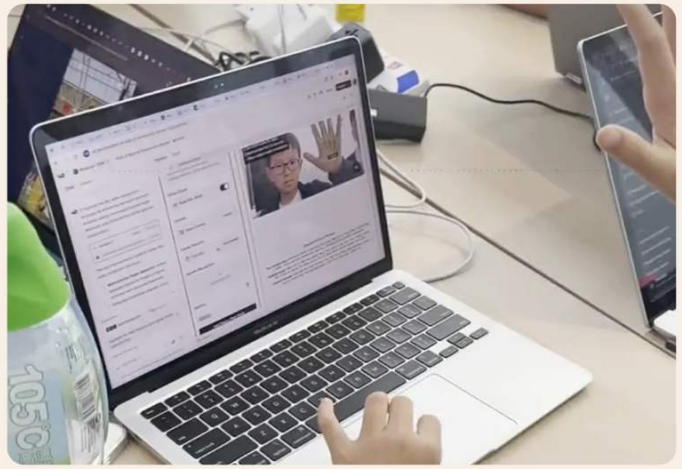
Demo day

Live pitches to judges and families.

10 Pitch and sell

WHERE IT STARTED · BODLEY ACADEMY, HONG KONG, SUMMER 2025

In five days, an 11-year-old *turned sign language into speech.*



Built in five days by a seventh grader

01 **Agency** FIND A PROBLEM	He chose sign language to speech because his cousin is deaf. Nobody assigned it.
02 **Taste** REAL-USER FEEDBACK	He kept improving it after the demo, because his grandmother couldn't use the first version yet.
03 **Craft** FINISH AND LAUNCH	The hardest part wasn't the AI. It was audio clear enough for her to trust.

AI tools did the building. *He did the choosing, judging, and finishing.*

20 students, each on their own problem

5 days, from the first idea to demo day

9.5/10 how likely students were to recommend it

THE CASE · DEMO DAY

On day five, teams pitched working products *to judges from Harvard, Stanford, Y Combinator, and AI labs.*



Judges test each product before scoring



Every team pitched its own product

JUDGED BY FOUNDERS AND OPERATORS FROM HARVARD Stanford Y Combinator *and AI labs*

PART 2

It worked because students made every decision, *and AI helped them build faster.*

Used well, AI deepens the thinking instead of replacing it.

THE RESEARCH · TWO 2025 STUDIES

Used as a shortcut, AI does the thinking.
Set up as a coach, it helps students learn more.

−17%

HIGH SCHOOL MATH · NEARLY 1,000 STUDENTS

exam grades after practicing with plain ChatGPT. A tutor version that gave hints, not answers, largely avoided the drop.

2×

HARVARD PHYSICS · A RANDOMIZED TRIAL

the learning gains of an active-learning class, in less time, with an AI tutor built on the same methods

Bastani and others, PNAS, June 25, 2025: grades fell 17% once plain ChatGPT was taken away; with safeguards, the harm was “largely mitigated”; Kestin and others, Scientific Reports, June 3, 2025: 194 Harvard physics students, median learning gains more than double

The ISB week gave each skill *a moment students couldn't skip.*

01 Agency

Choosing what is worth doing, before anyone assigns it.

HOW THE WEEK TRAINED IT

On day one, each student chooses the problem and pitches it in three minutes, with no slides.

02 Taste

Judging whether the output is actually good.

HOW THE WEEK TRAINED IT

By midweek, a prototype is online at a public link, and real users and mentors react every day.

03 Craft

Finishing the work so people can trust it.

HOW THE WEEK TRAINED IT

Building stops at a fixed hour each day, and on demo day students explain every decision live.

IS USING AI JUST CHEATING?

No. AI tools write the code, *and students make every decision.*

AI TOOLS AND MENTORS HANDLE

Writing the code

Putting it online

Fixing technical errors

STUDENTS DECIDE

Which problem, and for whom

What to cut after users try it

The name, the price, and the pitch

Your role becomes the coach, *and three questions do the coaching.*

THE TEACHER'S ROLE

From the person with the answers

↓

to the coach who asks the questions

It's what good teachers already do, and what our mentors did all week.

01
AGENCY

"Why this problem?"

They choose, not the AI.

02
TASTE

"How do you know it's good?"

They judge what it made.

03
CRAFT

"What would make someone trust it?"

They check it and finish it.

PART 3

Now we build one together. *No coding needed.*

Name a school chore. We build the fix together, live.

1 The case
What students built

2 Why it worked
The design

3 Build one, live
With the room

4 Take it back
What it takes

LIVE BUILD · 30 MINUTES

You bring a real school problem.
We build the fix together, live.

STEP 1 · YOU NAME IT

A chore or a snag

A workflow you repeat, or a problem you keep running into.

- *"I rewrite the same parent email every week."*
- *"Tracking which students turned in drafts."*
- *"Making one reading at three levels."*

→

STEP 2 · WE BUILD IT LIVE

Claude Code writes it

An AI tool that builds web apps from plain English.

Claude Code *by Anthropic*

```
› Make a page where I paste a reading and get it at three levels.
```

→ A working web app, in minutes

→

STEP 3 · WE ALL ITERATE

Open it, then improve it

We try it together and fix what's off, one plain sentence at a time.

- *"Make the buttons bigger."*
- *"Add a column for due dates."*
- *"Use simpler words for grade 6."*

PART 4

Take it back to your school: *teachers need AI too.*

What it takes to run this: teacher training and student programs.

WHY TEACHERS NEED AI TOO · EEF · GALLUP

AI gives teachers back time *to spend on their students.*

31% less time on lesson planning with AI in a randomized trial, with no drop in quality

5.9 hours saved a week by teachers who use AI weekly, about six weeks a school year

FROM A TEACHER WE SPOKE WITH

A grading timer, *not a grader*



An English teacher built a simple timer. It grades nothing: he still reads every paper and writes the feedback himself.

It gives every paper the same share of his time, and keeps his marking within a set limit.

AI handles the housekeeping. *The teacher keeps the judgment.*

EEF/NFER, randomized trial, 259 teachers in 68 schools (2024). Gallup and Walton Family Foundation (2025): weekly AI users, n=2,232, self-reported estimate.

Tools teachers build *for their own classrooms.*

ENGLISH · FROM A TEACHER WE SPOKE WITH

Poem notes, one click away

Shall I compare thee to a summer's day?

1. The question the whole sonnet answers
2. Beautiful, but it fades

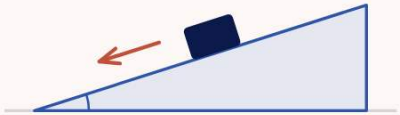
Instead of annotating a poem live while the class watches, the teacher builds the notes beforehand.

In class, one click highlights each line with its note.

Class time goes to discussing the poem.

SCIENCE · WHAT A STEM TEACHER CAN BUILD

A simulation made for this class



Angle **20°**

Friction **0.3**

Students take a model apart and rebuild it, or change one variable in a physics setup.

Built for this lesson, not the nearest YouTube demo.

Students test ideas instead of watching one.

TAKE IT BACK · PLANNING CHECKLIST

Your school already owns most of this.
Arena School can help with the rest.

YOUR SCHOOL OWNS

The learning goals and lasting skills

A five-day block on the calendar

Teachers, IT, and families on board

How students show their work

WHERE ARENA SCHOOL HELPS

Matching AI tools to each task

Prompting that builds thinking

Tools that work on your network

An idea to a product in five days

Checklist and China tool guide: arenaschool.org/talks/2026-10-acamis-tech-workshop

How we teach: arenaschool.org/approach

ARENA SCHOOL
www.arenaschool.org

Going further: we help schools bring AI *into teaching and learning.*

AI for Educators

Hands-on training at your school, with your teachers' own lessons

Student programs

A five-day build week like ISB's, co-run with your teachers

Curriculum design

Where AI fits in your courses, and how to assess it

SELECTED WORK

arenaschool.org/results

INTERNATIONAL SCHOOL OF BEIJING 北京顺义国际学校

Bodley Academy

清華大學 Tsinghua University

IF YOU REMEMBER ONE THING

AI doesn't have to do the thinking for students.

The design is up to us.

1

Students can ship

Live products in five days,
with no coding background.

2

Design decides

Students choose,
judge, and finish.

3

Start small, start now

One project, one class,
and three questions.

You don't need to code. You already know how to coach.

When you're ready to run your own, Arena School can help. arenaschool.org

Thank you! *We welcome your questions.*

SLIDES AND RESOURCES

arenaschool.org/talks/2026-10-acamis-tech-workshop

ARENA SCHOOL

BRING THIS TO YOUR SCHOOL

Book a call with us

About teacher training, student programs, or curriculum design.

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BOOK A CALL

arenaschool.org/contact